

German dictionary for AOSP keyboards

A German main dictionary in the AOSP binary `.dict` format. It works with [HeliBoard](#) and other keyboards based on the AOSP LatinIME (OpenBoard, ...), both for suggestions and for the system spell checker.

It combines word frequencies from real-world text with the Hunspell German dictionary to filter out misspellings and junk.

Building

Requirements: GNU Make (4.3 or newer), `python3`, `hunspell` and a Java runtime (`java`).

```
make
```

This produces:

File	Use for keyboard language
<code>build/main_de.dict</code>	German (<code>de</code>)
<code>build/main_de_DE.dict</code>	German (Germany) (<code>de-DE</code>)

Both files contain the same words and differ only in the locale written to their header.

The intermediate files `build/main_*.combined` (the text form of the dictionary) and `build/rejected_by_hunspell.txt` (every word that was dropped, with its count) are kept for inspection. `make clean` removes them and keeps the dictionaries; `make distclean` removes all of `build/`.

Set `SOURCE_DATE_EPOCH` to get a reproducible header date.

Installing in HeliBoard

Settings → Dictionaries → *Add dictionary from file*, then pick the `.dict` file that matches the keyboard language you actually use.

HeliBoard keeps a separate dictionary for each locale. A dictionary imported for German (`de`) is **not** used by a German (Germany) (`de-DE`) keyboard, which keeps using HeliBoard's built-in dictionary. Check the language shown in the import dialog; you can change it there with *Select language*.

For the spell checker, choose HeliBoard under Android Settings → System → Languages → Spell checker.

How the dictionary is made

`build_de_dict.py` does the following:

1. Reads the frequency list (`de/de_full_frequency.txt`) and keeps purely alphabetic Latin-1 words.
2. **Runs every word through Hunspell** (`de/de_DE.dic` / `de/de_DE.aff`) in lowercase, capitalized and (for short words) uppercase form. Only forms Hunspell accepts are kept, and the correct casing is restored, so nouns end up capitalized. This rules out misspellings, typos, names, and other bad or non-German words from the subtitle corpus. A capitalized form is only accepted if Hunspell derives it from a capitalized stem.
3. Adds the words in `de/allowlist.txt` . These are common colloquial words that Hunspell rejects, such as *ok*, *hey* and *naja*.
4. Maps word counts to AOSP frequencies 1–255 on a logarithmic scale. Nominalized infinitives (*das Essen*) get a lower frequency than the verb (*essen*).
5. Writes the AOSP "combined" wordlist format. `make` then compiles it to a binary dictionary with `dicttool_aosp makedict` .

Sources and credits

- `dicttool_aosp` , `dicttool_aosp.jar` were built from the [Android Open Source Project](#) source code by running `mm` in the `dicttool` package (`packages/inputmethods/LatinIME/tools/dicttool`). They are included here so the dictionary can be built without an AOSP checkout. They are licensed under the Apache License 2.0.
- `de/de_DE.dic` , `de/de_DE.aff` were taken from the Fedora `hunspell-de` package (version 20240224). They are based on [igerman98](#) by Björn Jacke, with the *frami* extension by Franz Michael Baumann, and are licensed under GPL-2.0 or GPL-3.0.
- `de/de_full_frequency.txt` is `content/2016/de/de_full.txt` from [FrequencyWords](#) by Hermit Dave. It contains word frequencies built from [OpenSubtitles](#) and is licensed under CC BY-SA 4.0. Many thanks for making this list available!

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